Indian Used Car Price Prediction

Executed project report with saved notebook outputs, tables, and charts

Project Description

Indian Used Car Price Prediction

Project Summary

This project is a clean, student-friendly implementation of a used car price prediction. The main goal is to estimate resale price from Indian used-car listings. The notebook keeps the workflow simple: load the dataset, understand the columns, clean the data, train a baseline model, and check how well the model performs.

No external repository links or profile links are included in this description. Everything needed to understand and run the project is kept inside this folder.

Problem Statement

A raw dataset is useful only when it is converted into a clear decision-making workflow. In this project, the data is processed step by step so that important patterns can be found and a machine learning model can be trained. The expected output is based on Price.

Files Included

File	Purpose	Quick Note
usedCars.csv	Main data source for this folder	1064 sampled rows x 19 columns
Notebook file	Complete Python workflow	Includes loading, cleaning, training, and evaluation
PDF report	Human-readable project summary	Matches the updated project structure

Important Columns

Column	Role
Id	Input feature used in the modelling workflow
Company	Input feature used in the modelling workflow
Model	Input feature used in the modelling workflow
Variant	Input feature used in the modelling workflow
FuelType	Input feature used in the modelling workflow
Colour	Input feature used in the modelling workflow
Kilometer	Input feature used in the modelling workflow
BodyStyle	Input feature used in the modelling workflow
TransmissionType	Input feature used in the modelling workflow
ManufactureDate	Input feature used in the modelling workflow
ModelYear	Input feature used in the modelling workflow
CngKit	Input feature used in the modelling workflow

Workflow Followed

1. Load the data from the local dataset file present in the same project folder.

2. Inspect the structure using shape, column names, missing values, and basic statistics.

3. Clean the dataset by removing empty columns, handling missing values, and keeping only useful features.

4. Prepare the model input by separating features and the target column.

5. Train a baseline model using a simple scikit-learn pipeline with preprocessing.

6. Evaluate the result using practical metrics such as accuracy, classification report, MAE, RMSE, or R2 score depending on the target type.

7. Write observations in a short and direct way so the project can be explained easily.

How to Run

1. Open the notebook in Jupyter Notebook, JupyterLab, Google Colab, or VS Code.

2. Keep the dataset files in the same folder as the notebook.

3. Run the cells from top to bottom.

4. Read the printed metrics and notes at the end before making conclusions.

Final Note

This version keeps the logic understandable and avoids unnecessary complexity. The aim is not to make the notebook look overloaded, but to make it readable, explainable, and useful for a student-level data science portfolio.

Executed Notebook Outputs

The outputs below were generated from the saved notebook after running the code cells in this project folder.

Cell 2 - 2. Load the dataset

```
Dataset shape: (1064, 19)
```

```
Id Company Model Variant FuelType \
0 555675 MARUTI SUZUKI CELERIO(2017-2019) 1.0 ZXI AMT O PETROL
1 556383 MARUTI SUZUKI ALTO LXI PETROL
2 556422 HYUNDAI GRAND I10 1.2 KAPPA ASTA PETROL
3 556771 TATA NEXON XT PLUS PETROL
4 559619 FORD FIGO EXI DURATORQ 1.4 DIESEL

Colour Kilometer BodyStyle TransmissionType ManufactureDate ModelYear \
0 Silver 33011 HATCHBACK NaN 2018-02-01 2019
1 Red 10187 HATCHBACK Manual 2021-03-01 2021
2 Grey 38183 HATCHBACK Manual 2015-03-01 2015
3 A Blue 13130 HATCHBACK NaN 2020-08-01 2019
4 Silver 104670 HATCHBACK Manual 2010-11-01 2009

CngKit Price Owner DealerState DealerName \
0 NaN 5.75 Lakhs 1st Owner Karnataka Top Gear Cars
1 NaN 4.35 Lakhs 1st Owner Karnataka Renew 4 u Automobiles PVT Ltd
2 NaN 4.7 Lakhs 1st Owner Karnataka Anant Cars Auto Pvt Ltd
3 NaN 9.9 Lakhs 1st Owner Karnataka Adeep Motors
4 NaN 2.7 Lakhs 2nd Owner Karnataka Zippy Automart

City Warranty QualityScore
0 Bangalore 1 7.6865
1 Bangalore 1 8.3365
2 Bangalore 1 7.9502
3 Bangalore 1 8.0162
4 Bangalore 0 7.7087
```

Cell 3 - 3. Understand the data

```
Rows: 1064
Columns: 19
```

```
column dtype missing_values unique_values
0 Id int64 0 1064
1 Company object 0 23
2 Model object 0 218
3 Variant object 0 575
4 FuelType object 1 5
5 Colour object 0 76
6 Kilometer int64 0 1055
7 BodyStyle object 0 10
8 TransmissionType object 714 9
9 ManufactureDate object 0 162
10 ModelYear int64 0 19
11 CngKit object 1042 2
12 Price object 0 367
13 Owner object 0 4
14 DealerState object 0 10
15 DealerName object 0 57
16 City object 0 11
17 Warranty int64 0 2
18 QualityScore float64 0 1036
```

Cell 4 - 4. Clean the data

```
Shape after simple cleaning: (1064, 19)
```

```
Id Company Model Variant FuelType \
0 555675 MARUTI SUZUKI CELERIO(2017-2019) 1.0 ZXI AMT O PETROL
1 556383 MARUTI SUZUKI ALTO LXI PETROL
2 556422 HYUNDAI GRAND I10 1.2 KAPPA ASTA PETROL
3 556771 TATA NEXON XT PLUS PETROL
4 559619 FORD FIGO EXI DURATORQ 1.4 DIESEL

Colour Kilometer BodyStyle TransmissionType ManufactureDate ModelYear \
0 Silver 33011 HATCHBACK NaN 2018-02-01 2019
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2 Grey 38183 HATCHBACK Manual 2015-03-01 2015
3 A Blue 13130 HATCHBACK NaN 2020-08-01 2019
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CngKit Price Owner DealerState DealerName \
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4 NaN 2.7 Lakhs 2nd Owner Karnataka Zippy Automart

City Warranty QualityScore
0 Bangalore 1 7.6865
1 Bangalore 1 8.3365
2 Bangalore 1 7.9502
3 Bangalore 1 8.0162
4 Bangalore 0 7.7087
```

Cell 5 - 5. Select the target column

```
Selected target column: Price
```

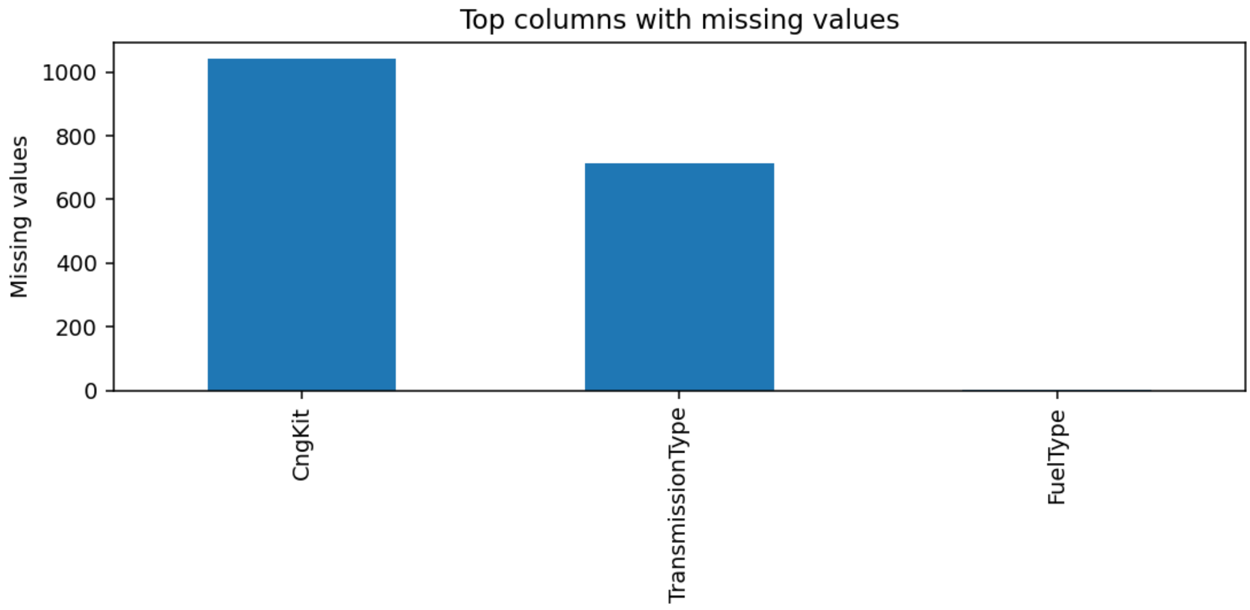
```
0 5.75 Lakhs
1 4.35 Lakhs
2 4.7 Lakhs
3 9.9 Lakhs
4 2.7 Lakhs
Name: Price, dtype: object
```

Cell 6 - 6. Quick EDA

```
Numeric columns: 5
Categorical columns: 14
```

```
count mean std min 25% \
Id 1064.0 568156.542293 16438.139974 525978.0 555321.500000
Kilometer 1064.0 52804.554511 33860.830048 454.0 32155.750000
ModelYear 1064.0 2016.890977 3.074392 2003.0 2015.000000
Warranty 1064.0 0.738722 0.439538 0.0 0.000000
QualityScore 1064.0 7.770687 0.737940 0.0 7.484075

50% 75% max
Id 572753.00000 583072.7500 589122.0000
Kilometer 49510.00000 68773.7500 639699.0000
ModelYear 2017.00000 2019.0000 2024.0000
Warranty 1.00000 1.0000 1.0000
QualityScore 7.82815 8.1483 9.5349
```



Cell 7 - 7. Prepare features and target

```
Dropped ID-like columns: ['Id', 'Kilometer', 'QualityScore']
Problem type: classification
Feature shape: (1064, 15)
Target unique values: 367
```

Cell 8 - 8. Train a baseline model

```
Training completed.
```

Cell 9 - 9. Evaluate the model

```
Accuracy: 0.0000

Classification report:
precision recall f1-score support

1 Lakhs 0.00 0.00 0.00 1.0
1.67 Lakhs 0.00 0.00 0.00 1.0
1.75 Lakhs 0.00 0.00 0.00 1.0
1.85 Lakhs 0.00 0.00 0.00 0.0
10 Lakhs 0.00 0.00 0.00 2.0
10.2 Lakhs 0.00 0.00 0.00 1.0
10.25 Lakhs 0.00 0.00 0.00 3.0
10.47 Lakhs 0.00 0.00 0.00 0.0
10.5 Lakhs 0.00 0.00 0.00 1.0
10.75 Lakhs 0.00 0.00 0.00 2.0
10.85 Lakhs 0.00 0.00 0.00 1.0
10.9 Lakhs 0.00 0.00 0.00 1.0
11 Lakhs 0.00 0.00 0.00 1.0
11.25 Lakhs 0.00 0.00 0.00 2.0
11.35 Lakhs 0.00 0.00 0.00 1.0
11.45 Lakhs 0.00 0.00 0.00 0.0
11.5 Lakhs 0.00 0.00 0.00 2.0
```

```
11.81 Lakhs 0.00 0.00 0.00 1.0
12.5 Lakhs 0.00 0.00 0.00 3.0
12.77 Lakhs 0.00 0.00 0.00 1.0
12.95 Lakhs 0.00 0.00 0.00 1.0
13 Lakhs 0.00 0.00 0.00 1.0
13.25 Lakhs 0.00 0.00 0.00 1.0
13.35 Lakhs 0.00 0.00 0.00 1.0
14.25 Lakhs 0.00 0.00 0.00 1.0
14.75 Lakhs 0.00 0.00 0.00 3.0
14.9 Lakhs 0.00 0.00 0.00 1.0
14.91 Lakhs 0.00 0.00 0.00 0.0
15.5 Lakhs 0.00 0.00 0.00 2.0
15.75 Lakhs 0.00 0.00 0.00 1.0
16.5 Lakhs 0.00 0.00 0.00 2.0
16.8 Lakhs 0.00 0.00 0.00 1.0
17.51 Lakhs 0.00 0.00 0.00 1.0
18 Lakhs 0.00 0.00 0.00 1.0
18.00 Lakhs 0.00 0.00 0.00 1.0
18.25 Lakhs 0.00 0.00 0.00 1.0
18.5 Lakhs 0.00 0.00 0.00 1.0
18.9 Lakhs 0.00 0.00 0.00 2.0
19.4 Lakhs 0.00 0.00 0.00 0.0
19.5 Lakhs 0.00 0.00 0.00 1.0
2.1 Lakhs 0.00 0.00 0.00 0.0
2.25 Lakhs 0.00 0.00 0.00 1.0
2.35 Lakhs 0.00 0.00 0.00 1.0
2.4 Lakhs 0.00 0.00 0.00 1.0
2.5 Lakhs 0.00 0.00 0.00 1.0
2.75 Lakhs 0.00 0.00 0.00 2.0
2.8 Lakhs 0.00 0.00 0.00 1.0
2.85 Lakhs 0.00 0.00 0.00 1.0
2.95 Lakhs 0.00 0.00 0.00 1.0
20 Lakhs 0.00 0.00 0.00 0.0
21 Lakhs 0.00 0.00 0.00 1.0
22.5 Lakhs 0.00 0.00 0.00 0.0
23.5 Lakhs 0.00 0.00 0.00 1.0
25 Lakhs 0.00 0.00 0.00 1.0
25.9 Lakhs 0.00 0.00 0.00 0.0
26.75 Lakhs 0.00 0.00 0.00 2.0
3.18 Lakhs 0.00 0.00 0.00 0.0
3.2 Lakhs 0.00 0.00 0.00 1.0
3.25 Lakhs 0.00 0.00 0.00 1.0
3.3 Lakhs 0.00 0.00 0.00 2.0
3.4 Lakhs 0.00 0.00 0.00 1.0
3.45 Lakhs 0.00 0.00 0.00 1.0
3.5 Lakhs 0.00 0.00 0.00 4.0
3.8 Lakhs 0.00 0.00 0.00 1.0
3.9 Lakhs 0.00 0.00 0.00 1.0
3.95 Lakhs 0.00 0.00 0.00 2.0
31.75 Lakhs 0.00 0.00 0.00 1.0
4 Lakhs 0.00 0.00 0.00 3.0
4.05 Lakhs 0.00 0.00 0.00 0.0
4.1 Lakhs 0.00 0.00 0.00 1.0
4.25 Lakhs 0.00 0.00 0.00 0.0
4.29 Lakhs 0.00 0.00 0.00 1.0
4.3 Lakhs 0.00 0.00 0.00 1.0
4.35 Lakhs 0.00 0.00 0.00 2.0
4.37 Lakhs 0.00 0.00 0.00 0.0
4.45 Lakhs 0.00 0.00 0.00 2.0
4.49 Lakhs 0.00 0.00 0.00 1.0
4.5 Lakhs 0.00 0.00 0.00 3.0
4.55 Lakhs 0.00 0.00 0.00 1.0
4.7 Lakhs 0.00 0.00 0.00 1.0
4.71 Lakhs 0.00 0.00 0.00 1.0
4.75 Lakhs 0.00 0.00 0.00 2.0
4.8 Lakhs 0.00 0.00 0.00 3.0
4.85 Lakhs 0.00 0.00 0.00 2.0
4.95 Lakhs 0.00 0.00 0.00 2.0
5 Lakhs 0.00 0.00 0.00 1.0
5.00 Lakhs 0.00 0.00 0.00 1.0
5.1 Lakhs 0.00 0.00 0.00 2.0
5.15 Lakhs 0.00 0.00 0.00 1.0
5.25 Lakhs 0.00 0.00 0.00 1.0
5.29 Lakhs 0.00 0.00 0.00 0.0
5.3 Lakhs 0.00 0.00 0.00 3.0
5.35 Lakhs 0.00 0.00 0.00 1.0
5.36 Lakhs 0.00 0.00 0.00 1.0
5.4 Lakhs 0.00 0.00 0.00 1.0
5.45 Lakhs 0.00 0.00 0.00 2.0
5.49 Lakhs 0.00 0.00 0.00 1.0
5.5 Lakhs 0.00 0.00 0.00 8.0
5.6 Lakhs 0.00 0.00 0.00 1.0
5.68 Lakhs 0.00 0.00 0.00 1.0
5.75 Lakhs 0.00 0.00 0.00 1.0
5.8 Lakhs 0.00 0.00 0.00 1.0
5.85 Lakhs 0.00 0.00 0.00 1.0
5.87 Lakhs 0.00 0.00 0.00 1.0
5.9 Lakhs 0.00 0.00 0.00 1.0
5.95 Lakhs 0.00 0.00 0.00 2.0
6 Lakhs 0.00 0.00 0.00 2.0
6.1 Lakhs 0.00 0.00 0.00 1.0
6.15 Lakhs 0.00 0.00 0.00 1.0
6.2 Lakhs 0.00 0.00 0.00 1.0
```

```
6.25 Lakhs 0.00 0.00 0.00 6.0
6.35 Lakhs 0.00 0.00 0.00 1.0
6.5 Lakhs 0.00 0.00 0.00 4.0
6.55 Lakhs 0.00 0.00 0.00 1.0
6.7 Lakhs 0.00 0.00 0.00 1.0
6.75 Lakhs 0.00 0.00 0.00 5.0
6.9 Lakhs 0.00 0.00 0.00 2.0
6.93 Lakhs 0.00 0.00 0.00 1.0
6.95 Lakhs 0.00 0.00 0.00 4.0
7 Lakhs 0.00 0.00 0.00 1.0
7.15 Lakhs 0.00 0.00 0.00 1.0
7.19 Lakhs 0.00 0.00 0.00 1.0
7.25 Lakhs 0.00 0.00 0.00 2.0
7.29 Lakhs 0.00 0.00 0.00 1.0
7.3 Lakhs 0.00 0.00 0.00 2.0
7.4 Lakhs 0.00 0.00 0.00 1.0
7.43 Lakhs 0.00 0.00 0.00 1.0
7.49 Lakhs 0.00 0.00 0.00 1.0
7.5 Lakhs 0.00 0.00 0.00 6.0
7.6 Lakhs 0.00 0.00 0.00 1.0
7.65 Lakhs 0.00 0.00 0.00 1.0
7.8 Lakhs 0.00 0.00 0.00 1.0
7.85 Lakhs 0.00 0.00 0.00 2.0
7.9 Lakhs 0.00 0.00 0.00 1.0
7.95 Lakhs 0.00 0.00 0.00 1.0
7.99 Lakhs 0.00 0.00 0.00 1.0
8 Lakhs 0.00 0.00 0.00 1.0
8.25 Lakhs 0.00 0.00 0.00 2.0
8.35 Lakhs 0.00 0.00 0.00 1.0
8.5 Lakhs 0.00 0.00 0.00 1.0
8.7 Lakhs 0.00 0.00 0.00 1.0
8.75 Lakhs 0.00 0.00 0.00 1.0
8.95 Lakhs 0.00 0.00 0.00 1.0
9.16 Lakhs 0.00 0.00 0.00 1.0
9.21 Lakhs 0.00 0.00 0.00 1.0
9.25 Lakhs 0.00 0.00 0.00 2.0
9.45 Lakhs 0.00 0.00 0.00 1.0
9.5 Lakhs 0.00 0.00 0.00 3.0
9.75 Lakhs 0.00 0.00 0.00 4.0
9.9 Lakhs 0.00 0.00 0.00 2.0
9.92 Lakhs 0.00 0.00 0.00 1.0

accuracy 0.00 213.0
macro avg 0.00 0.00 0.00 213.0
weighted avg 0.00 0.00 0.00 213.0
```